Young's Modulus Non – Uniform Bending

Aim

To find the Young's modulus of the given material bar by non-uniform bending using pin and microscope method.

Apparatus

Pin and Microscope arrangement, Scale, Verniercallipers, Screw gauge, Weight hanger, Material bar or rod.

Theory

Young's modulus is named after Thomas Young, 19thcentury, British scientist. In solid mechanics, Young's modulus is defines as the ratio of the longitudinal stress over longitudinal strain, in the range of elasticity the Hook's law holds (stress is directly proportional to strain). It is a measure of stiffness of elastic material.

If a wire of length L and area of cross-section 'a' be stretched by a force F and if a change (increase) of length 'l' is produced, then

$$Young's\ modulus = \frac{Normal\ stress}{Longitudinal\ strain} = \frac{F/a}{l/L}$$

Non Uniform Bending Using Pin and Microscope

Here the given beam (meter scale) is supported symmetrically on two knife edges and loaded at its centre. The maximum depression is produced at its centre. Since the load is applied only one point of the beam, the bending is not uniform throughout the beam and the bending of the beam is called non- uniform bending.

In non-uniform bending (central loading), the Young's modulus of the material of the bar is given by

$$Y = \frac{mgl^3}{48Ie} \qquad (1)$$

I is the moment of inertia of the bar.

For a rectangular bar,

$$I = \frac{bd^3}{12}$$

Substituting (4) in (3)

In non-uniform bending, the young's modulus of the material of the bar is given by,

$$Y = \frac{mgl^3}{4bd^3e} \qquad (2)$$

m - Mass loaded for depression.

g - Acceleration due to gravity.

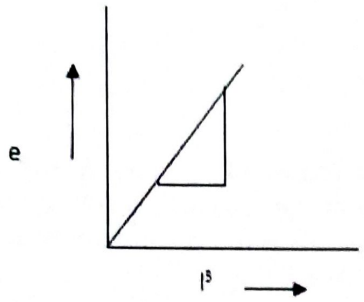
l - Length between knife edges.

b - Breadth of the bar using Verniercallipers.

d - Thickness of the bar using screw gauge.

e - Depression of the bar.

Graph between depression and length



From graph $\frac{l^3}{e}$ can be calculated.

Procedure

Non-Uniform Bending

The given bar is supported symmetrically on two knife edges. The length l of the bar between the knife edges is measured. A weight hanger is suspended exactly at the midpoint of the bar. A pin is fixed vertically at the midpoint of the bar. A pin is fixed vertically at the midpoint of the bar with its pointed end upwards. The microscope is arranged in front of the pin and focused at the tip of the pin. The slotted weights are added one by one on both the weight hangers and removed one by one a number of times, so that the bar is brought into an elastic mood. With the some "dead load" W_0 on each weight hanger, the microscope is adjusted so that the image of the tip of the pin coincides with the point of intersection of cross wires. The reading of the vernier scale and vernier of microscope are taken. Weights are added one by one and corresponding reading are taken. From these readings, the mean depression (e) of the mid-point of the bar for a given mass is determined. From the microscope reading, the mean depression (e) for a given mass is found. $\frac{l^3}{e}$ The value of is calculated and hence calculate the young's modulus of the given material bar.

Observations and Calculations of Non-Uniform Bending

Value of 1 m.s.d = 1/20
Number of divisions on the Vernier, n = 50
Least count of microscope = 1 m.s.d/n = 1/1000 = 0.001 cm

No	Distance of the knife edges , l (cm)	Load M(kg)	Telescope reading			depression for load 4m, e (cm)	Mean e (cm)	$\frac{l^3}{e}$ (cm^3)	Mean $\frac{l^3}{e}$ (cm^3)
			Loading (cm)	unloading (cm)	mean (cm)				
1		Wo Wo+m Wo+2m Wo+3m Wo+4m Wo+5m Wo+6m Wo+7m			Xo X1 X2 X3 X4 X5 X6 X7	X4-X0 X5-X1 X6-X2 X7-X3			
2									
3									
4									

Thickness of the material bar "d" = ... mm.

Breadth of the material bar "b" = ...cm.

Mean value of l^3/e = ...m.

Load applied for depression "e" = ... m.

Young's modulus of the material bar, $Y = \frac{mgl^3}{4bd^3e}$ = ...N/m^2.

Result

1. Young's modulus of the given material using non uniform bending method =.....Nm^{-2}.